Tobin Wainer

twainer@stsci.edu
https://tobin-wainer.github.io

Professional Employment:

Post-Doctoral Researcher at the Space Telescope Science Institute
Advisor: Kristen McQuinn — September 2026 - Present

Pre-Doctoral Fellow at the Center for Computational Astrophysics
Advisors: Julianne Dalcanton, Mike Grudić — September 2024 - January 2025

Post-Baccalaureate Researcher at the University of Utah
Advisor: Gail Zasowski — June 2021 - August 2022

Education

Doctorate of Philosophy in Astronomy — July 2026
University of Washington, Seattle, Washington

- Thesis: "Bridging the Scales of Star Formation: Embedded Clusters to Galactic Histories"
- Advisors: Ben Williams, Julianne Dalcanton, James Davenport

Masters of Science in Astronomy — June 2024
University of Washington, Seattle, Washington

Honors Bachelor of Science in Physics — May 2021
Astronomy Emphasis
University of Utah, Salt Lake City, Utah

- Unweighted GPA: 3.90/4.0 (3.84/4.0 Physics GPA)
- Honors Thesis; "Star Clusters in the Triangulum Galaxy: Star Cluster Catalog and Mass Function Fitting"
- Advisors: Anil Seth, Cliff Johnson
- Undergraduate Research Scholar Designation

Grants

Science PI: JWST Cycle 6, *Title removed to remain dual anonymous* — Pending

Co-II JWST Cycle 6, 5 Proposals — Pending

Co-I: JWST Cycle 5, ID. #9622, *A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33* — $37,000

Science PI: TESS Cycle 7, *Searching for Stellar Activity Cycles Using Flares Identified is TESS Lightcurves* — $70,000

Co-I: HST Proposal. Cycle 32, ID. #17833, *Bringing HST to the VLA: The Interaction of Stars and Gas in the Local Group*

Co-I: JWST Proposal. Cycle 3, ID. #5145, *Resolving the Global Stellar Populations and Star Formation History of the Starburst Galaxy M82*

Science PI: Astrophysics Data Analysis Program (ADAP), *Constraining the Ages of Unresolved Star Clusters with Ensemble TESS Light Curves* — $262,000

University of Utah Undergraduate Research Opportunity Program Grant - 2020
$1200

Honors and Awards

University of Washington Astronomy Department Graduate Student Prize for Excellence in Research	2024
Predoctoral fellow at the Center for Computational Astrophysics	2024
Honorable Mentions in the NSF Graduate Research Fellowship Program	2024
University of Utah Physics Department Most Outstanding Teaching Assistant Award	2022
Honorable Mentions in the NSF Graduate Research Fellowship Program	2022
Honorable Mentions in the Chambliss Achievement Student Award competition for undergraduates at the 237th AAS conference	2021
University of Utah Outstanding Undergraduate Research Award in Astronomy and Astrophysics	2021
University of Utah Representative for Research on Capitol Hill; presenting research to state legislators	2021
Made Dean's List each eligible semester of enrollment	F17, S18, F19, F20
Research Experience for Undergraduates at Northwestern University	2020
Summer Undergraduate Research Program at the University of Utah	2019
One-year Peer Advisor Scholarship for dedication to student learning and excellence in teaching at the University of Utah	$2000, 2019

First Author Publications

Wainer, T., [4 Authors], 2026, "Searching for Stellar Activity Cycles using Flares II: The TESS CVZ", *Submitted to AAS Journals*

Wainer, T., [21 Authors], 2026, "The Panchromatic Hubble Andromeda Southern Treasury (PHAST). II. The Spatially Resolved Recent Star Formation History in M31", *The Astrophysical Journal, 2026ApJ Volume 1006, id.195W*

Wainer, T., [8 Authors], 2026, "The Timescales of Embedded Star Formation as Observed in STARFORGE", *The Astrophysical Journal, 2026ApJ Volume 998, id.215W*

Wainer, T., [17 Authors], 2025, "Crowded Field Photometry with Rubin: Exploring 47 Tucanae with Data Preview 1", *Research Notes of the AAS, Volume 9, Issue 7, id.171*

Wainer, T., [4 Authors], 2024, "Searching for Stellar Activity Cycles using Flares: The Short and Long Timescale Activity Variations of TIC-272272592", *The Astronomical Journal, 2024AJ, 168, 232W*

Wainer, T., [17 Authors], 2024, "The Panchromatic Hubble Andromeda Treasury: Triangulum Extended Region (PHATTER). VI. The High-mass Stellar Initial Mass Function of M33", *The Astronomical Journal, 2024AJ, 168, 86W*

Wainer, T., [3 Authors], 2023, "ELK: A python package for correcting, analyzing, and diagnosing TESS integrated light curves", *The Journal of Open Source*

Software 2023 JOSS, 8(90), 05605

Wainer, T., [9 Authors], 2023, "Catalog of Integrated-light Star Cluster Light Curves in TESS", *The Astronomical Journal, 2023AJ, 166, 106W*

Wainer, T., [12 Authors], 2022, "The Panchromatic Hubble Andromeda Treasury: Triangulum Extended Region (PHATTER). III. The Mass Function of Young Star Clusters in M33", *The Astrophysical Journal, 2022ApJ, 928, 15W*

Selected Contribution Publications
*** Indicates Student Paper**

Lazzarini, M., [...] **Wainer, T.**, 2026, "Optical Spectroscopy of a Candidate O-star X-Ray Binary in M33", *The Astrophysical Journal, 2026ApJ, Volume 1010, id.43*

Tran, D., [...] **Wainer, T.**, 2026, "Automated Computer Vision Cluster Identification in the Fireworks Galaxy", *The Astrophysical Journal, 2026ApJ, Volume 1006, id.238*

Wang, Y., [...] **Wainer, T.**, 2025, "Optical Counterparts to X-ray sources in LSST DP1", *The Astrophysical Journal Supplement Series, 2026ApJS, Volume 284, id.13*

Chen, Z., [...] **Wainer, T.**, 2025, "PHAST. The Panchromatic Hubble Andromeda Southern Treasury. I. Ultraviolet and Optical Photometry of over 90 Million Stars in M31", *The Astrophysical Journal, 979, 35c*

* Gollotti, G., [...] **Wainer, T.**, 2024, "Two Intriguing Transits of HIP 23309 Observed by TESS", *Research Notes of the AAS, 8, 293*

Peltonen, J., [...] **Wainer, T.**, 2024, "JWST reveals star formation across a spiral arm in M33", *Monthly Notices of the Royal Astronomical Society, 527, I4*

Peltonen, J., [...] **Wainer, T.**, 2023, "Clusters, Clouds, and Correlations: Relating Young Clusters to Giant Molecular Clouds in M33 and M31", *Monthly Notices of the Royal Astronomical Society, 522, I4*

Johnson, L.C., **Wainer, T.**, [11 Authors], 2022, "The Panchromatic Hubble Andromeda Treasury: Triangulum Extended Region (PHATTER). IV. M33 Star Cluster Catalog", *The Astrophysical Journal 2022ApJ, 938, 81J*

Teaching Experience

Course Instructor September 2025 - December 2025
Yakima Valley Community College

- *Introduction to Scientific Computing*
  - Developed course syllabus and content
  - Led weekly lectures of 10 students

Teaching Assistant September 2022 - June 2026
University of Washington

- *Astronomy 101a, 101b; Introduction to Astronomy In person and Online*
- *Astronomy 192; Pre-Major in Astronomy and Physics*
  - Led 3 weekly sections of 30 students each
  - Made weekly recap lecture videos

Teaching Assistant August 2018 - May 2022
University of Utah

- *Astronomy 1010, 4090; The Universe, Stellar Astrophysics*
  - Helped facilitate weekly 5-person teaching staff meeting to plan for the next week
  - Facilitated 20-person group sessions and provided one-on-one support
  - Collaborated with professor to design class structure
- *Physics 2210, 2020; Physics I for Scientists and Engineers, General Physics II*
  - Collaborated with teachers and program directors to meet academic needs of students
  - Graded student assignments and exams
- *LEAP 1010, 1020; Pre-Law I, Pre-Law II, Role of Law in Society*
  - Led class when professors were attending conferences

Students Mentored

Current:

- Graduate Students: Charlie Willard (2025-), Ethan Potter (2026-), Emerald Gingell-Molina (2026-)
- Undergraduate Students: Holland Stuart (2025-)

Former:

- Undergraduate Students: Giovanni Gollotti (2024-2025), Danbi Kim(2023), Amelia Ross (2023)

Presentations and Conference Contributions

Space Telescope Science Institute Colloquium, Invited Talk — October 2026
Baltimore, Maryland

Cool Stars 23, Contributed Poster (Link to Abstract) — June 2026
Tokyo, Japan

PHANGS Star Clusters Group Meeting, Invited Talk — October 2025
Virtual

Center for Computational Astrophysics Pre-Doc Symposium, Invited Talk (Link to talk) — June 2025
New York, New York

Ballyfest; Star Formation, Stellar Feedback, and the Ecology of Galaxies, Contributed Poster — May 2025
Visegrad, Hungary

2025 Spring Symposium: Inter+Stellar: Harnessing the Intersection Between Stars and the Interstellar Medium, Contributed Poster — May 2025
Baltimore, Maryland

TESS Science Conference III, Contributed Poster — August 2024
Cambridge, Massachusets

243rd Meeting of the AAS, Contributed Talk — January 2024
New Orleans, Louisiana

8th TESS Asteroseismic Science Consortium, Contributed Talk — July 2023
Honolulu, Hawaii

240th Meeting of the AAS, Poster 201.01 (Link to Abstract) June 2022
Pasadena, California

Undergraduate Research Symposium, Contributed Poster April 2021
University of Utah, Salt Lake City, Utah

237 Meeting of the AAS, Poster 150.07 (Link to Abstract) January 2021
Virtual

Research Experience for Undergraduate Symposium, Contributed Poster August 2020
Northwestern University and Adler Planetarium, Chicago, Illinois

235 Meeting of the AAS, Poster 306.02 (Link to Abstract) January 2020
Honolulu, Hawaii

2019 Physics Congress, Contributed Poster November 2019
Providence, Rhode Island

Summer Research Symposium, Contributed Poster August 2019
University of Utah, Salt Lake City, Utah

Service

Apache Point Observatory Telescope Allocation Committee May 2023 - January 2026

- University of Washington graduate student representative for allocating telescope time of the 3.5 meter APO.
- Tasks include reading proposals and collaborating with faculty members of the TAC, to equitably allocate time.

Astronomy on Tap Roadshow Co-Founder September 2024 - June 2026

- Bring Astronomy on tap to rural communities across Washington State.

Astronomy on Tap Seattle Technical Assistant September 2023 - June 2026

- In charge of all technical aspects of the show from microphones and projection, to live streaming the show.

Senator in the University of Washington Graduate and Professional Student Senate September 2023 - September 2025

- Serve the broader University student population on issues like affordability and student wellness.

University of Washington Astronomy Planetarium Technical Assistant September 2023 - 2024

- Point person for technological issues with the planetarium.
- Perform routine service to Planetarium software.

Society of Physics Students Communications Director May 2019 - May 2021
University of Utah Chapter

- Raised over $7500 to send 12 undergraduates to the quadrennial SPS conference in Rhode Island
- Orchestrated monthly outreach events for local high school and middle school students

- Coordinated lab tours to help undergraduates discover research opportunities

Undergraduate Student Advisory Committee Associate Chair August 2019 - May 2021
University of Utah Department of Physics and Astronomy

- Worked with a six-person committee to generate a report providing a recommendation about a professor's Retention, Promotion, or Tenure
- Conducted interviews with professors, graduate students, and undergraduates to gather information
- Coordinate with the committee chair to facilitate meetings

Other Experience and Activities

Research on Capitol Hill February 2021
Salt Lake City Capital Building

- Selected as 1 of 15 undergraduates to represent the University of Utah
- Presented research to state legislators

Pro Bono Law Clinic Volunteer August 2019 - December 2019
Salt Lake City Pro Bono Law Clinic

- Worked alongside law students and practicing attorneys to provide clients with legal advice
- Conducted client interviews and consulted with attorneys
- Helped direct clients through necessary paperwork

Telescope Time Awarded

3 Nights- Apache Point Observatory 3.5M: Fall 2024

8 Half Nights- CHIRON echelle spectrograph, SMARTS 1.5 M telescope Fall 2025

Skills

- Programming Languages: Python, HTML
- Astronomy Software: DS9, Astropy, MATCH, SKIRT, SLUG
- Data Analysis: integrated spectroscopy, using large survey data (HST, JWST), HST, JWST image processing, time series analysis using TESS
- Technical skills: Stellar population synthesis modeling
- Statistical Techniques: Bayesian Modelling, Regression, Bootstrap sampling
- Other software: LATEX, Github

Professional Memberships

American Astronomical Society (AAS)
American Physics Society (APS)
Society Of Physics Students (SPS)

Hobbies

- Backpacking
- Mountaineering
- Skiing
- Sports fanatic